

→ Mohammed Asem Rashed

→ 05B

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Answer the que no-1

① Brittleness & Ductility:

Brittleness: Brittleness is the property of a material to break suddenly with very little or no plastic deformation.

Ex: Glass, Cast Iron, ceramics

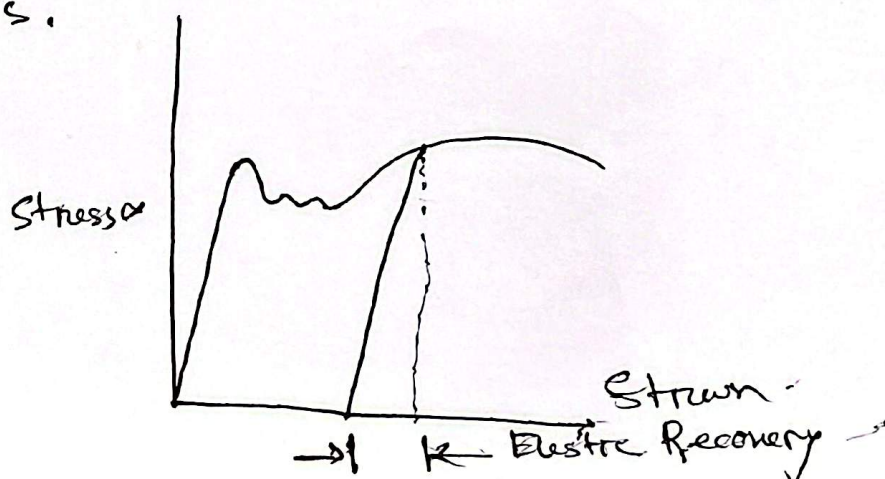
Ductility: Ductility is the property of material to undergo large plastic deformation before fracture. It can be drawn into wires.

Ex: Mild Steel, copper, Aluminium.

② Mild Steel Tensile test Observation:

When the load is removed before the yield point, the material behaves elastically and returns to its original length because only elastic deformation occurs.

When the load is removed after the yield point, the material undergoes plastic deformation. Therefore, it cannot return to its original length. It remains as it is.



Ans to the Que No-2

a) Strength of Material and types

Strength is the ability of a material to resist applied load without failure or excessive deformation.

- Tensile Strength
- Compressive Strength
- Shear Strength
- Torsional Strength

b) DT vs NDT

<u>Destructive Testing</u>	<u>Non-Destructive Testing</u>
① Specimen is damaged during testing	① Specimen remains undamaged
② Measures Actual strength	② Detects defect without damage
③ Tensile Test	③ Ultrasonic Test, or RT

c) Purpose of Material Testing

- To determine mechanical properties of material
- To ensure quality and safety
- To select suitable material for engineering applications

Ans to Que-no-3

a) Charpy impact test vs Izod Impact test

<u>Charpy test</u>	<u>Izod Test</u>
① Specimen is placed horizontally	① Specimen is placed vertically
② Support at both ends	② Fixed at one end,
③ Hammer strikes opposite notch at the center,	③ Hammer strike above the notch

b) Numerical

* cross-section Area

$$A = \frac{\pi d^2}{4} = \frac{\pi (87)^2}{4} = 5944.68 \text{ mm}^2$$

$$\text{* Breaking Strength} = \frac{24200}{5944.68}$$

$$= 40.71 \text{ N/mm}^2$$

Given :

$$Diameter = 87 \text{ mm}$$

$$L_0 = 207 \text{ mm}$$

$$L_f = 256 \text{ mm}$$

$$P = 242 = 24200 \text{ N}$$

$$\text{* percentage} = \frac{L_f - L_0}{L_0} \times 100$$

$$= \frac{256 - 207}{207} \times 100$$

$$= 23.67 \%$$

Answer